ASSIGNMENT-09

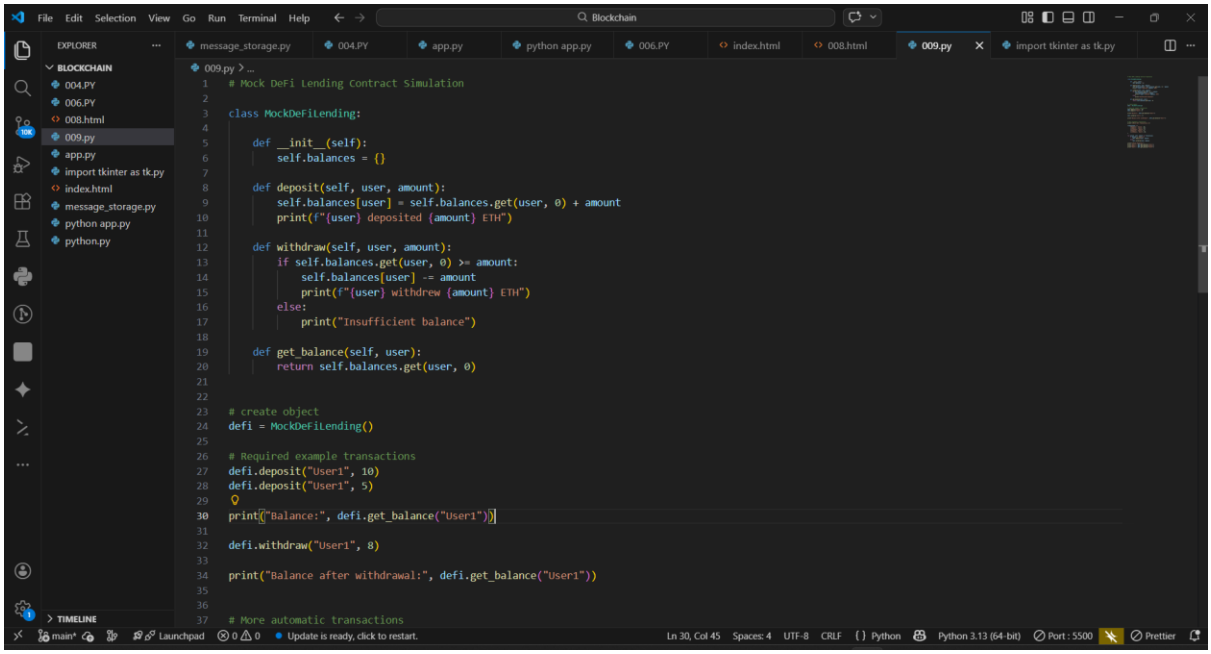
ROLL NO:2303A51360

BATCH NO:29

QUESTION:

To interact with a decentralized finance (DeFi) protocol on a blockchain testnet by writing and executing a smart contract that deposits and withdraws tokens, demonstrating basic DeFi operations without real funds.

CODE:



```python
# Mock DeFi Lending Contract Simulation

class MockDeFiLending:

    def __init__(self):
        self.balances = {}

    def deposit(self, user, amount):
        self.balances[user] = self.balances.get(user, 0) + amount
        print(f"{user} deposited {amount} ETH")

    def withdraw(self, user, amount):
        if self.balances.get(user, 0) >= amount:
            self.balances[user] -= amount
            print(f"{user} withdrew {amount} ETH")
        else:
            print("Insufficient balance")

    def get_balance(self, user):
        return self.balances.get(user, 0)


# create object
defi = MockDeFiLending()

# Required example transactions
defi.deposit("User1", 10)
defi.deposit("User1", 5)

print("Balance:", defi.get_balance("User1"))

defi.withdraw("User1", 8)

print("Balance after withdrawal:", defi.get_balance("User1"))


# More automatic transactions
```

```python
transactions = [
    ("deposit", "User2", 20),
    ("deposit", "User3", 15),
    ("withdraw", "User2", 5),
    ("deposit", "User1", 3),
    ("withdraw", "User3", 10)
]

for action, user, amount in transactions:
    if action == "deposit":
        defi.deposit(user, amount)
    else:
        defi.withdraw(user, amount)

print("\nFinal Balances:")
```

```
User3: 5
PS C:\Users\nampa\OneDrive\Blockchain> & C:\Users\nampa\AppData\Local\Programs\Python\Python313\python.exe c:/Users/nampa/OneDrive/Blockchain/009.py
User1 deposited 10 ETH
User1 deposited 5 ETH
Balance: 15
User1 withdrew 8 ETH
Balance after withdrawal: 7

More DeFi Transactions:

User2 deposited 20 ETH
User3 deposited 15 ETH
User2 withdrew 5 ETH
User1 deposited 3 ETH
User3 withdrew 10 ETH

Final Balances:
User1: 10
User2: 15
User3: 5
PS C:\Users\nampa\OneDrive\Blockchain>
```